

BHARATH SAI LINGAMPALLI

Stony Brook, NY | +1 (934) 246-3276 | Email: bharathsai0203@gmail.com | [LinkedIn](#)

EDUCATION

Stony Brook University

Master of Science in Computer Science

Coursework: Data Science Fundamentals, Natural Language Processing, Database Systems, Machine Learning, Distributed Systems.

Stony Brook, NY, USA

August 2024 – Present

SKILLS

Programming Languages: Python, Java, Go, C++, TypeScript, SQL, Bash, C#, C, HTML, CSS, JavaScript, PHP

Frameworks: Spring Boot, FastAPI, Flask, React, Node.js, gRPC/REST, .NET, PyTorch, TensorFlow, Hugging Face, Scikit-learn, PySpark

Technologies: PostgreSQL, MongoDB, Redis, Kafka, ONNX Runtime, CUDA, FAISS, AWS (EC2/S3/RDS/Lambda), Microsoft Azure, Docker, Kubernetes, Terraform, Jenkins, NGINX, Prometheus, Grafana, Git, DynamoDB, Linux, CI/CD, Rest, RabbitMQ, Jenkins, Terraform

WORK EXPERIENCE

Stony Brook University

January 2025 – Present

Research Assistant | React.js, HTML, CSS, JavaScript, Chrome Extension APIs, Flask, Selenium, BeautifulSoup, NLP, JSON

- Created a Chrome Extension using React.js and **Python-based Selenium scrapers**, consolidating multi-source forum data into a unified JSON schema for cross-platform search, knowledge retrieval, and accessibility research.
- Applied NLP models with **GPU-accelerated preprocessing (CUDA & PyTorch)** to rank and surface high-quality responses, reducing cognitive load for blind and low-vision users by 60% while enhancing **ARIA**, semantic markup, and screen-reader support.
- Implemented and deployed REST APIs and fault-tolerant scraping pipelines integrating 5+ forums with **<1 s query latency**, featuring **parallelized ingestion**, **GPU-backed text vectorization**, and error recovery to ensure high reliability and scalability under load.

PayBis Club Private Limited

June 2023 – July 2024

Software Engineer | Java (Spring Boot), GO (Gin/Fiber), PostgreSQL, MongoDB, Redis, Kafka, AWS (EC2, RDS, S3), Docker, Kubernetes

- Developed and **scaled backend** services for credit-card applications, KYC verification, and transaction processing using **Java and Go microservices**, ensuring modularity, fault isolation, and resilience across distributed systems with containerized deployments.
- Designed and maintained **REST/gRPC APIs** for customer data, limits, and rewards using **PostgreSQL and MongoDB**, implementing **connection pooling and caching** to ensure data consistency and seamless integration between backend modules.
- Engineered and optimized **Kafka-based event pipelines** for asynchronous payment workflows, improving throughput, reducing message lag, and cutting failure rates by ~15 % during peak-load scenarios through better partitioning and consumer balancing.
- Enhanced performance and scalability with **Redis caching**, **Dockerized services**, and **Kubernetes orchestration on AWS**, leveraging auto-scaling and **Prometheus/Grafana** dashboards to reduce API latency by ~35% under high traffic.

PROJECTS

Concurrent-Safe Data Structures | C++, Multithreading

- Implemented high-performance concurrent data structures in **C++** using **multithreading** with fine-grained locking and atomic synchronization primitives, to ensure **thread safety and linear scalability** under high contention.
- Enhanced performance for **read-heavy workloads**, reducing access latency by **800 ms** on **1 million random reads** through **lock contention minimization**, **cache-efficient memory access** and load balanced thread scheduling.

Rental Management System | Spring Boot, Java, React.js, MongoDB, Redis, Docker, AWS EC2, Git

- Built secure **backend services** to handle property listings, accounts, and transactions with 99.9% data reliability and <200ms response times. Delivered a responsive frontend with **React.js** for an intuitive user experience.
- Integrated **geospatial queries** and **Redis caching**, improving property search speed by 45% and accuracy by 30% across datasets.
- Automated **CI/CD** pipelines on **AWS**, enabling zero-downtime deployments and cutting release cycle time by 40%.

DormMate — AI Helpdesk Assistant | Python, FastAPI, PyTorch (CUDA), ONNX Runtime, FAISS-GPU, React, Prometheus

- Built an asynchronous microservice pipeline with **GPU-accelerated intent, entity, and retrieval models**, achieving sub-500ms response latency through optimized batching, memory management, and **concurrent request handling**.
- Developed a **custom CUDA kernel** for batched cosine-similarity and Top-K reranking using **shared memory, warp-level reductions, and parallel execution**, delivering a **2.7× speed-up** over CPU baselines in **PyTorch extensions**.
- Integrated ONNX Runtime (CUDA EP) + FAISS-GPU with full observability via **Prometheus/Grafana monitoring, performance profiling, and load testing**, realizing a **4× throughput gain** and improved inference efficiency.

Distributed Key-Value Store using RAFT | C++, RAFT, gRPC, Docker

- Created a fault-tolerant store implementing the **RAFT consensus protocol** for leader election, log replication, and fault-tolerant consistency across cluster nodes, enabling recovery from node failures in **less than 3 seconds**.
- Integrated **consistent-hashing-based sharding** to support automated partition rebalancing, maintaining **balanced load distribution (less than 10 % variance)** during node joins and departures.
- Improved throughput by ~35% via **asynchronous replication** and tuned leader failover, improving overall **cluster efficiency**.